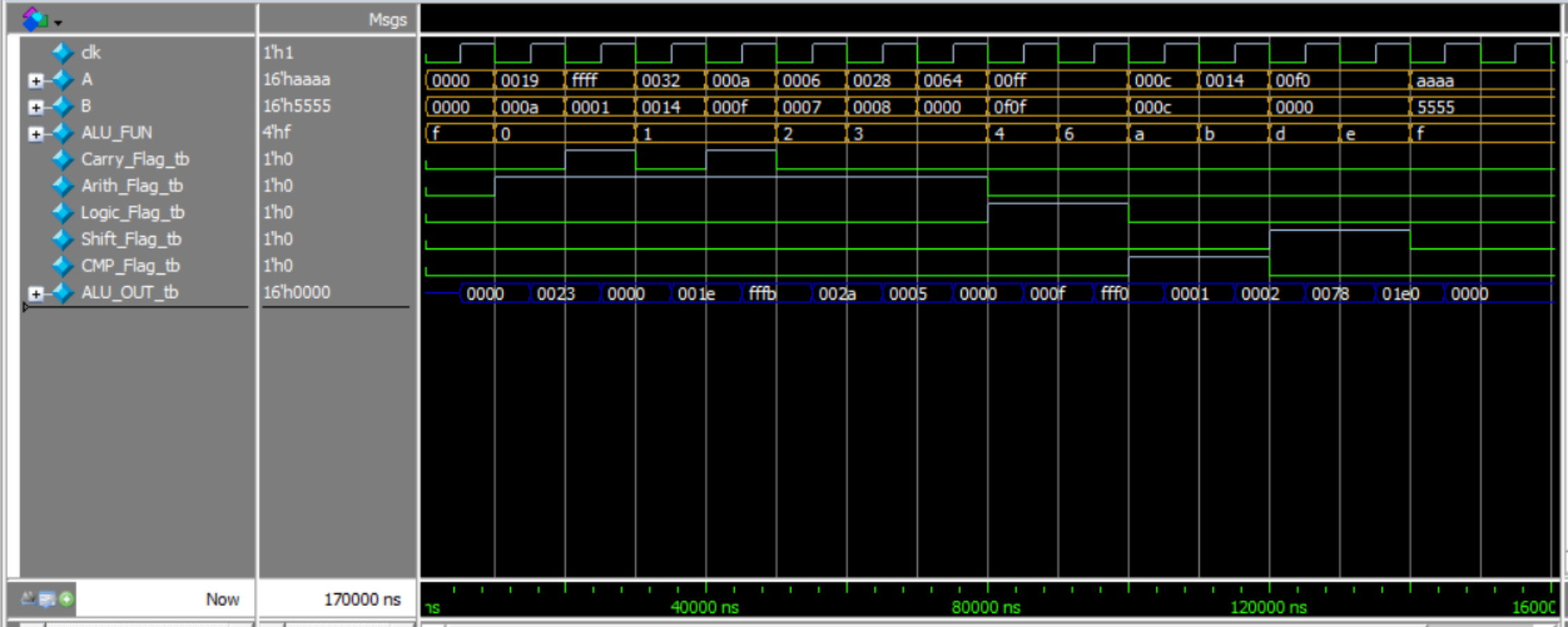

Msgs
clk
1'h1
A
16'haaaa
B
16'h5555
ALU_FUN
4'hf
Carry_Flag_tb
1'h0
Arith_Flag_tb
1'h0
Logic_Flag_tb
1'h0
Shift_Flag_tb
1'h0
CMP_Flag_tb
1'h0
ALU_OUT_tb
16'h0000
0000 0019 ffff 0032 000a 0006 0028 0064 00ff 000c 0014 00f0 aaaa
0000 000a 0001 0014 000f 0007 0008 0000 0f0f 000c 0000 5555
f 0 1 2 3 4 6 a b d e f
0000 0023 0000 001e fffb 002a 0005 0000 000f fff0 0001 0002 0078 01e0 0000
Now
170000 ns
ns
40000 ns
80000 ns
120000 ns
1600

```
# vsim -voptargs="+acc" work.ALU_16B_tb -coverage -sv_seed random -onfinish stop -l siml.log
# Start time: 16:28:07 on Jul 07,2026
# ** Note: (vsim-3813) Design is being optimized due to module recompilation...
# Loading work.ALU_16B_tb(fast)
# Loading work.ALU_16B(fast)
# Sv_Seed = 1384113629
# 0
# [PASS]                                          Normal Addition | ALU_OUT: 0x0023 (Dec: 35) | Flags (C,S,L,A,CMP): 00010 at time 20000
# [PASS]                                Addition with Overflow | ALU_OUT: 0x0000 (Dec: 0) | Flags (C,S,L,A,CMP): 10010 at time 30000
# [PASS]                                   Normal Subtraction | ALU_OUT: 0x001e (Dec: 30) | Flags (C,S,L,A,CMP): 00010 at time 40000
# [PASS]                               Subtraction with Borrow | ALU_OUT: 0xfffb (Dec: 65531) | Flags (C,S,L,A,CMP): 10010 at time 50000
# [PASS]                                 Normal Multiplication | ALU_OUT: 0x002a (Dec: 42) | Flags (C,S,L,A,CMP): 00010 at time 60000
# [PASS]                                     Normal Division | ALU_OUT: 0x0005 (Dec: 5) | Flags (C,S,L,A,CMP): 00010 at time 70000
# [PASS]                           Division By Zero Protection | ALU_OUT: 0x0000 (Dec: 0) | Flags (C,S,L,A,CMP): 00010 at time 80000
# [PASS]                                          Bitwise AND | ALU_OUT: 0x000f (Dec: 15) | Flags (C,S,L,A,CMP): 00100 at time 90000
# [PASS]                                         Bitwise NAND | ALU_OUT: 0xfff0 (Dec: 65520) | Flags (C,S,L,A,CMP): 00100 at time 100000
# [PASS]                              Comparison: A Equal to B | ALU_OUT: 0x0001 (Dec: 1) | Flags (C,S,L,A,CMP): 00001 at time 110000
# [PASS]                          Comparison: A Greater Than B | ALU_OUT: 0x0002 (Dec: 2) | Flags (C,S,L,A,CMP): 00001 at time 120000
# [PASS]                                          Shift Right | ALU_OUT: 0x0078 (Dec: 120) | Flags (C,S,L,A,CMP): 01000 at time 130000
# [PASS]                                           Shift Left | ALU_OUT: 0x01e0 (Dec: 480) | Flags (C,S,L,A,CMP): 01000 at time 140000
# [PASS]                          Unhandled ALU_FUN Protection | ALU_OUT: 0x0000 (Dec: 0) | Flags (C,S,L,A,CMP): 00000 at time 150000
# ====================================
#        ALL ALU TESTS COMPLETED
# ====================================
# ** Note: $finish    : ALU_16B_tb.v(177)
#    Time: 170 us  Iteration: 0  Instance: /ALU_16B_tb
# Break in Module ALU_16B_tb at ALU_16B_tb.v line 177
```